

Machine Learning – Chapter 4

Similarity-Based Learning | S. Sridhar
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Chapter 4: Similarity-Based Learning

Instance-based learning

Nearest-neighbor learning

Distance and similarity measures

Weighted K-Nearest-Neighbor (WKNN)

Nearest Centroid Classifier

Locally Weighted Regression (LWR)

Similarity / Instance-Based Learning

Stores or retains representative training instances.

Prediction is delayed until a query instance arrives.

Similarity between query and stored instances drives prediction.

Often called lazy learning because model construction is minimal.

Instance-Based vs Model-Based Learning

Instance-Based

- Retains training examples
- Prediction is computation-intensive
- Local decisions
- Flexible decision boundaries

Model-Based

- Learns compact parameters/rules
- Training is computation-intensive
- Global or parametric structure
- Faster prediction in many cases

Similarity Measures

Similarity quantifies how close two instances are.

Distance is often converted into similarity.

Choice of measure should reflect feature types and scale.

Feature normalization is important when units differ.

Euclidean Distance

$$d(x,z) = \sqrt{\sum_i (x_i - z_i)^2}$$

Common distance for continuous numeric features; sensitive to feature scale.

Manhattan Distance

$$d(x,z) = \sum_i |x_i - z_i|$$

Also called L1 distance; useful when movement along coordinate dimensions is meaningful.

Minkowski Distance

$$d(x,z) = (\sum_i |x_i - z_i|^p)^{1/p}$$

$p = 1$ gives Manhattan distance; $p = 2$ gives Euclidean distance.

Why Normalization Matters

Suppose Age ranges 18–60 and Income ranges 20,000–2,00,000.

Income can dominate Euclidean distance because of its larger numerical scale.

Min–max normalization or standardization puts features on comparable scales.

Distance-based algorithms are particularly sensitive to this issue.

Nearest-Neighbor Learning

Given a query point, calculate its distance to training instances.

Select the nearest instance(s).

For classification, use the class label(s) of selected neighbors.

For regression, aggregate their target values.

1-Nearest Neighbor

Find the single closest training instance.

Assign its class to the query for classification.

For regression, use the nearest target value.

Simple and highly local, but sensitive to noise and individual examples.

K-Nearest Neighbors

Select K closest training instances.

Classification: majority vote.

Regression: average target values.

K controls locality: small K gives flexible boundaries; larger K gives smoother decisions.

Choosing K

Small K: low bias, potentially high variance.

Large K: smoother model, potentially higher bias.

K is commonly selected using validation or cross-validation.

Odd K can reduce ties in binary classification.

KNN Classification Example

Query: a new point with two features.

Compute distances to all training points.

Sort distances and select K nearest points.

Count class labels among neighbors.

Assign the majority class.

Weighted K-Nearest Neighbor

Not all neighbors should contribute equally.

Closer neighbors receive larger weights.

A common choice is inverse-distance weighting.

The prediction is therefore influenced more strongly by nearby instances.

WKNN Weighting

$$w_i = 1 / (d_i + \varepsilon)$$

ε avoids division by zero. Other decreasing functions of distance may also be used.

WKNN Classification

$$\text{Score}(c) = \sum_i w_i \cdot I(y_i = c)$$

Choose the class with the largest weighted score.

WKNN Regression

$$\hat{y} = \sum_i w_i y_i / \sum_i w_i$$

A weighted average of target values, emphasizing nearby observations.

Worked WKNN Example

Suppose three neighbors have distances 1, 2 and 4.

Using $w = 1/d$ gives weights 1, 0.5 and 0.25.

The closest neighbor contributes four times as much as the farthest.

Calculate weighted class scores or a weighted numeric prediction.

Nearest Centroid Classifier

Compute the centroid of each class.

Measure distance from the query to each class centroid.

Assign the query to the nearest centroid.

Simple, fast and effective when classes are reasonably compact.

Class Centroid

$$\mu_c = (1/N_c) \sum x_i$$

The centroid is the mean feature vector of instances belonging to class c .

Nearest Centroid: Example

Class A centroid = mean of all A feature vectors.

Class B centroid = mean of all B feature vectors.

For a new point, calculate distance to both centroids.

Choose the class whose centroid is closer.

Locally Weighted Regression

LWR fits a regression model locally around a query point.

Nearby observations receive higher weights.

A separate local model is effectively constructed for each query.

Useful when the overall relationship is nonlinear but locally smooth.

LWR Weight Function

$$w_i(x_q) = \exp(-(x_q - x_i)^2 / 2\tau^2)$$

τ controls the width of the neighborhood; smaller τ gives stronger locality.

LWR Procedure

1. Choose a query point.
2. Compute distance from query to training points.
3. Assign distance-based weights.
4. Fit weighted regression using those weights.
5. Predict the query output.
6. Repeat for another query if required.

Role of Bandwidth τ

Small $\tau \rightarrow$ only very local points have strong influence.

Large $\tau \rightarrow$ more points contribute substantially.

Bandwidth controls the bias–variance trade-off.

Select τ using validation/cross-validation.

Advantages of Similarity-Based Learning

Simple conceptual model.

Naturally captures local patterns.

Flexible decision boundaries.

Can adapt as new training examples are added.

Useful when a meaningful similarity measure is available.

Limitations

Prediction can be computationally expensive.

Requires storing training instances.

Sensitive to irrelevant features and feature scales.

Choice of distance metric and K matters.

High-dimensional spaces can reduce the usefulness of distance measures.

Curse of Dimensionality

As dimensionality increases, data become sparse.

Distances between points can become less discriminative.

More data may be required to represent local neighborhoods adequately.

Feature selection or dimensionality reduction can help.

Comparison of Methods

KNN / WKNN

- Uses local neighbors
- Classification or regression
- K controls neighborhood
- WKNN gives closer points more influence

Nearest Centroid / LWR

- Centroid uses class summaries
- LWR fits local regression
- Both exploit locality differently
- Appropriate scaling remains important

Chapter 4: Key Takeaways

Similarity-based methods make predictions from nearby instances.

Distance metrics determine neighborhood structure.

KNN uses equal neighbor influence; WKNN uses distance-based influence.

Nearest Centroid compares queries with class prototypes.

LWR builds a locally weighted regression model.

Chapter 4: Examination Focus

Calculate Euclidean/Manhattan/Minkowski distance.

Solve KNN classification problems.

Solve WKNN using inverse-distance weights.

Compute class centroids and classify a query.

Explain LWR and the effect of bandwidth.

Discuss advantages, limitations and curse of dimensionality.